Library Management System (LMS)

Features and Architecture Overview

Library Management System Overview

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What it is

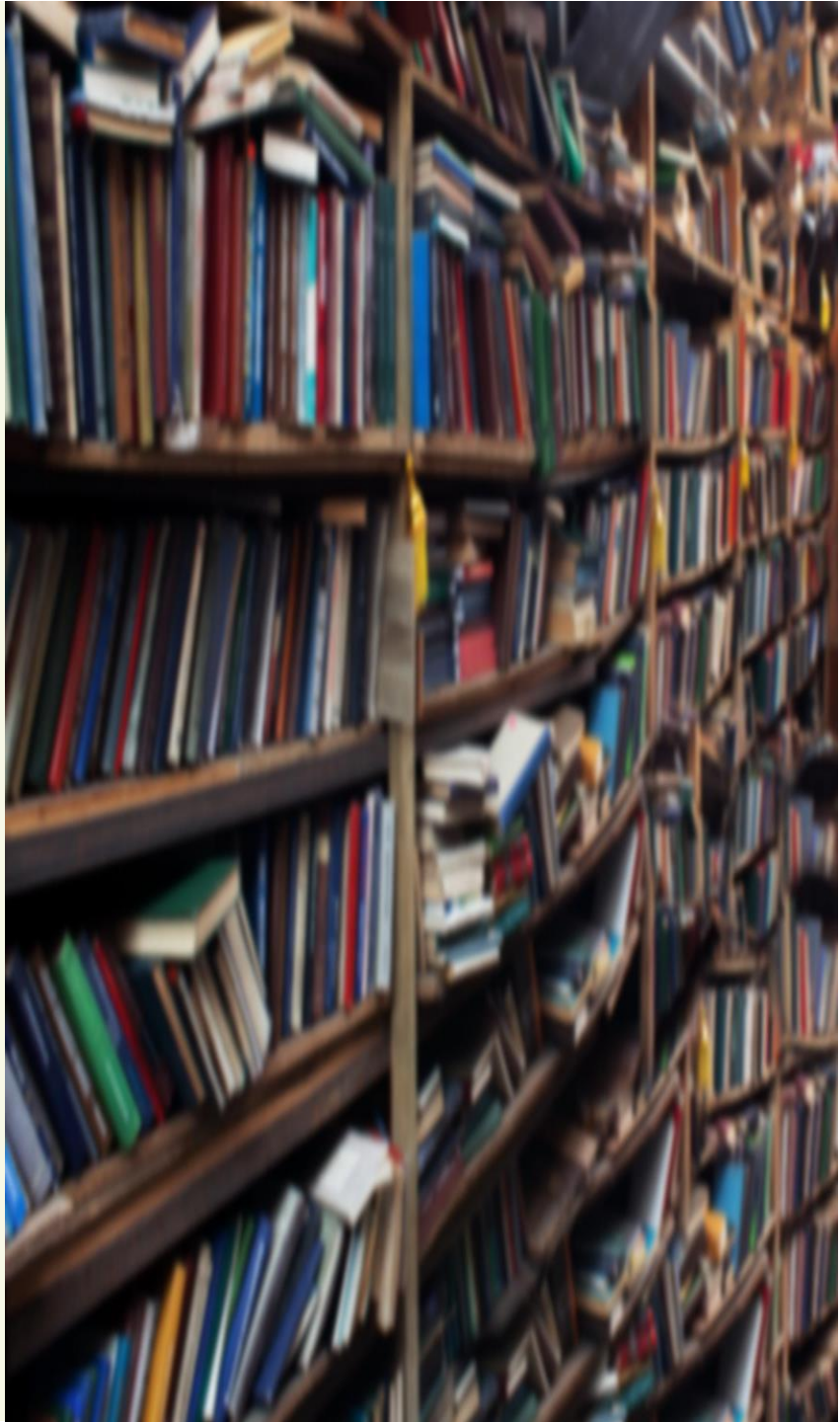
A web-based app that replaces manual record keeping with a centralized system to manage books, members, and loan records through an intuitive interface.

How it's built

Implemented with ASP.NET Core Razor Pages for clean, page-focused CRUD workflows and backed by SQL Server for reliable storage and verification.

What it demonstrates

Real-world library operations: availability tracking, issuing/returns, due dates, historical loan data, and auditable timestamps (CreatedAt, IssuedOn, DueOn, ReturnedOn).



Technology Stack

Technologies and Tools Used

Frontend (UI)

Razor Pages (.cshtml) with HTML5, CSS3, and Bootstrap for clean, responsive screens and light server-side rendering.

Backend (App Logic)

ASP.NET Core (.NET 8) with C# for routing, validation, and coordinating between the UI and the data layer.

Data Access & Database

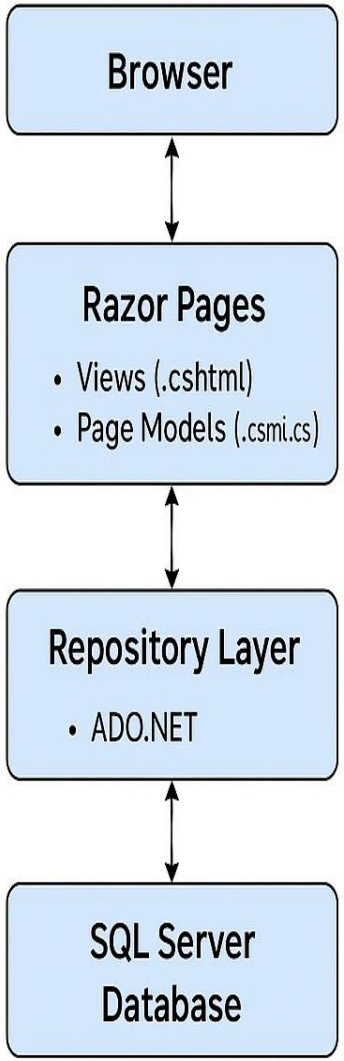
ADO.NET for explicit SQL control (queries, parameters, CRUD) with SQL Server storing books, members, and loans via keys and relationships.

Development Tools

Visual Studio Code for coding/debugging and SQL Server Management Studio for database inspection and verification.

Razor Pages Architecture and Flow

Library Management System Architecture



Razor Pages Architecture & Flow

Page-based MVC pattern

Each feature page is a pair: a Razor View (.cshtml) for rendering and a PageModel (.cshtml.cs) for request logic.

Request handling in PageModel

Methods like OnGet/OnPost process user actions, apply validation and business decisions, then call the repository layer.

Data flow and separation of concerns

SQL Server → repositories (ADO.NET + queries) → PageModels → Views, improving maintainability and debugging clarity.

Features Overview

Features Summary

Book Management

Maintains accurate book records, copies, and real-time availability.

Member Management

Registers users and keeps member details current and validated.

Loan Management

Issues and returns books with due dates and status tracking (active, returned, overdue).

Supports full CRUD, rule-based validation, and clear UI navigation with feedback messages.

Book Management Features

Book Management Features

Core catalog actions

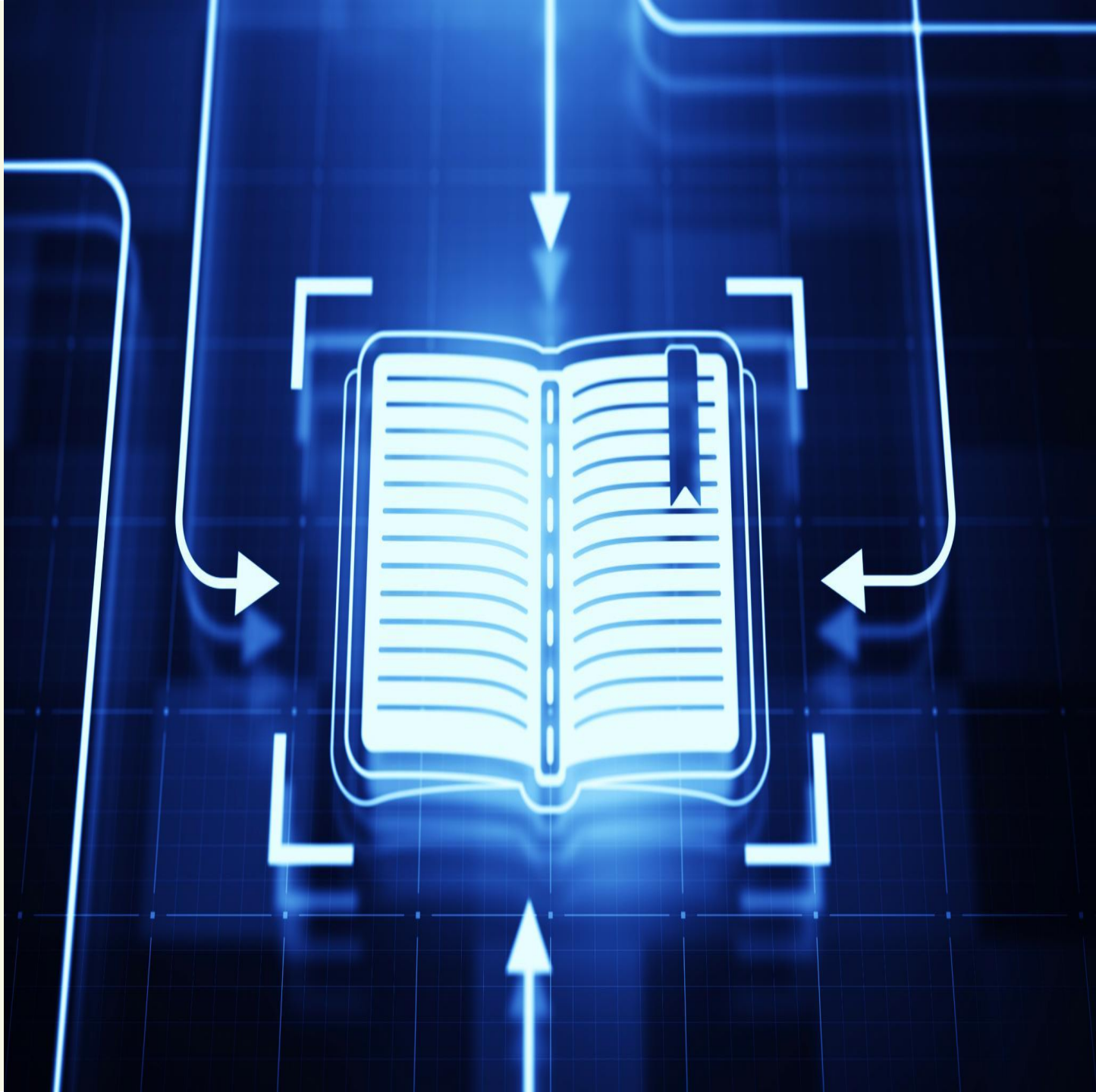
Add books with title, author, and total copies; edit records to update details; delete items removed from inventory.

Visibility and tracking

View all books in a table; when enabled, available copies update automatically as books are issued and returned.

Audit-ready timestamps

CreatedAt and UpdatedAt help track when records are added or modified, keeping catalog data accurate for reliable loans.



Member Management Features

Member Management Features

Register & Maintain Records

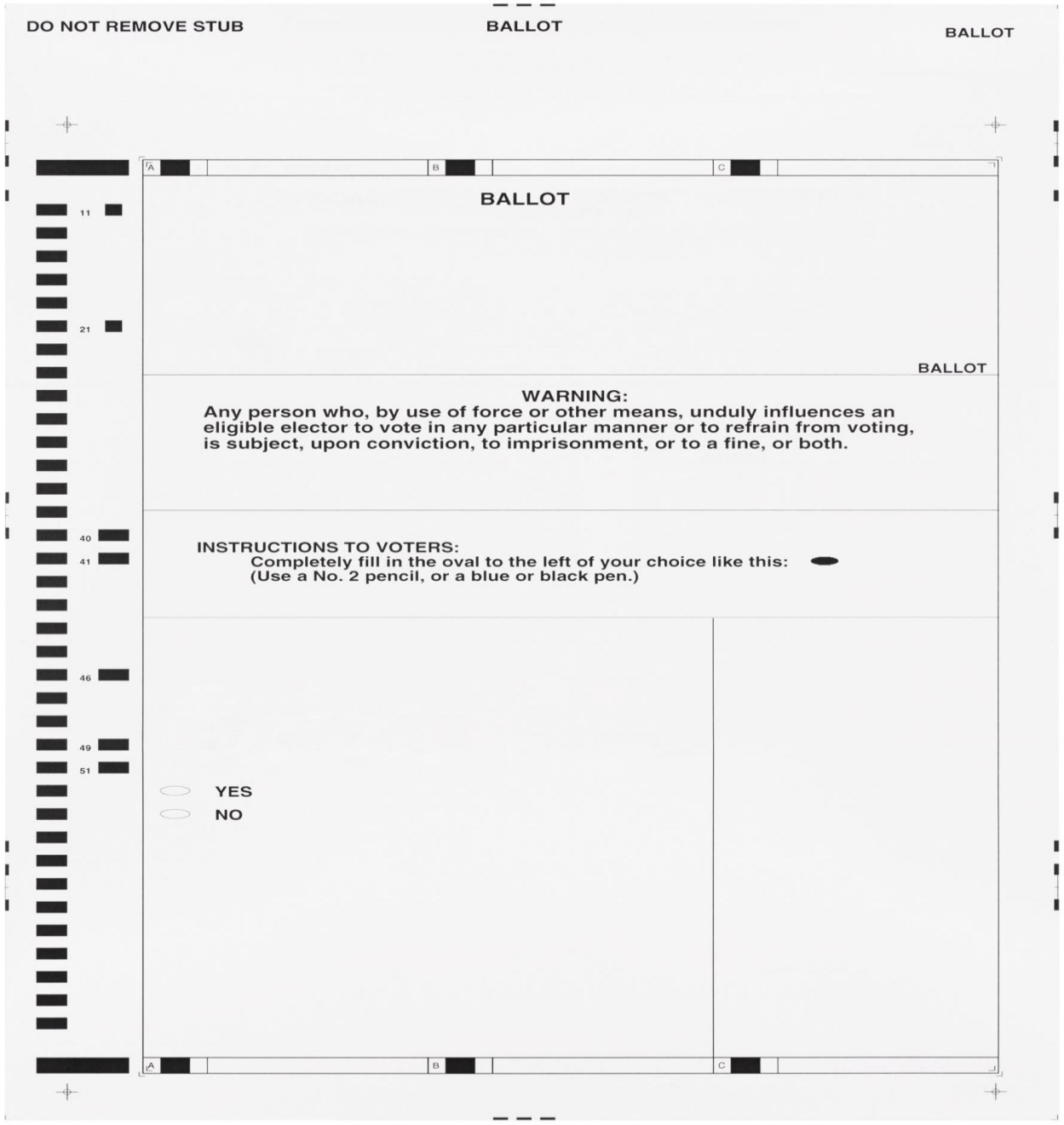
Capture name, email, phone, and address; update details anytime or remove members when needed.

List View for Fast Management

Members appear in a single list, making it easy to locate, review, and manage individual records.

Validation & Audit Trail

Basic required-field checks plus a CreatedAt timestamp to support accurate member-to-loan association.



Loans Module

Issue & Return Workflow

Issue a book

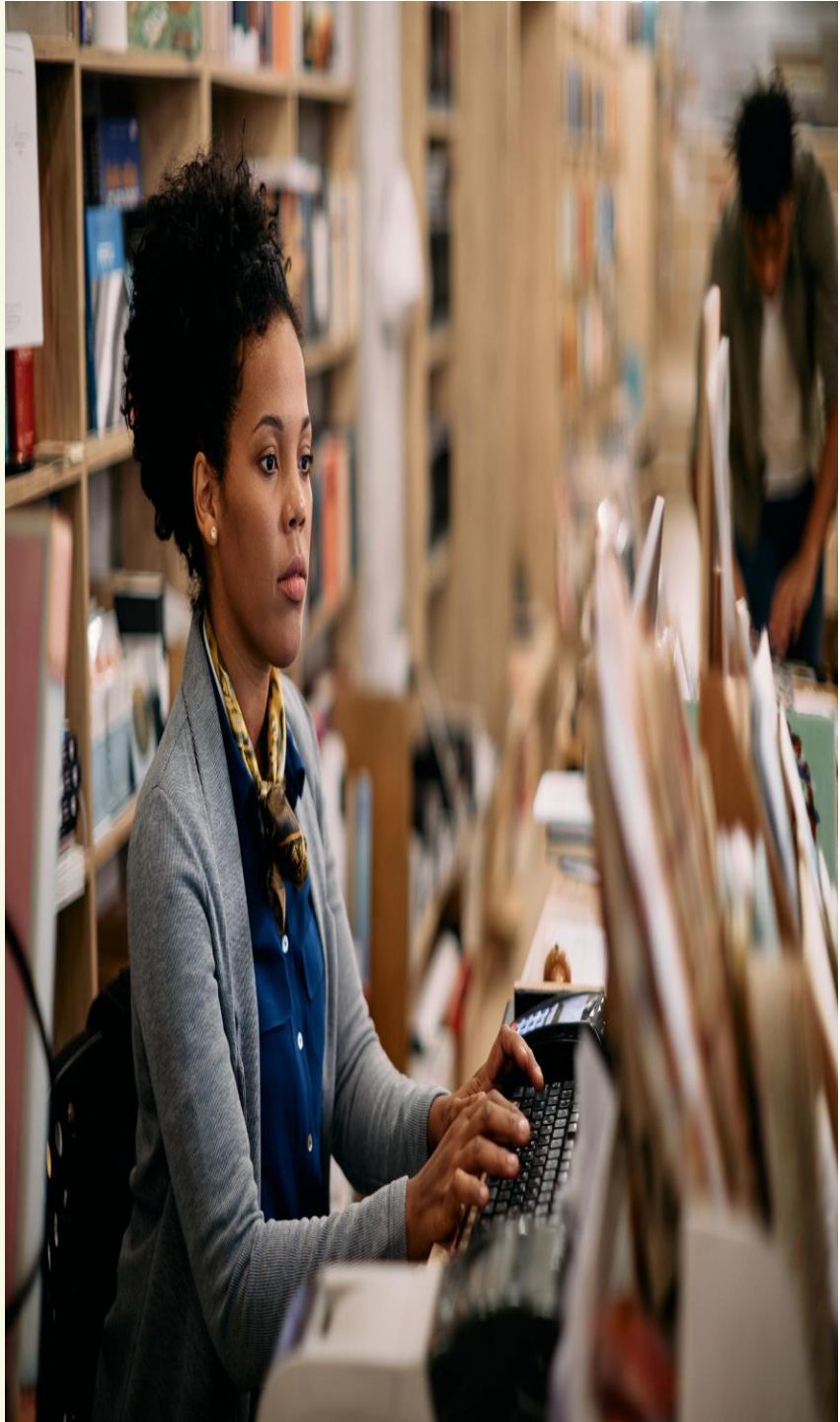
Select a member and a book, then choose a loan duration. The system sets IssuedOn to today and calculates DueOn from the duration.

Persist the loan

A new loan record is inserted with ReturnedOn set to null, enabling clear tracking of active vs. completed loans.

Return & review

On return, ReturnedOn is updated to today; availability counts adjust if enabled. View all loans to spot active, returned, and overdue items.



Rule-Based Business Logic

Rule-Based Business Logic

Core lending rules

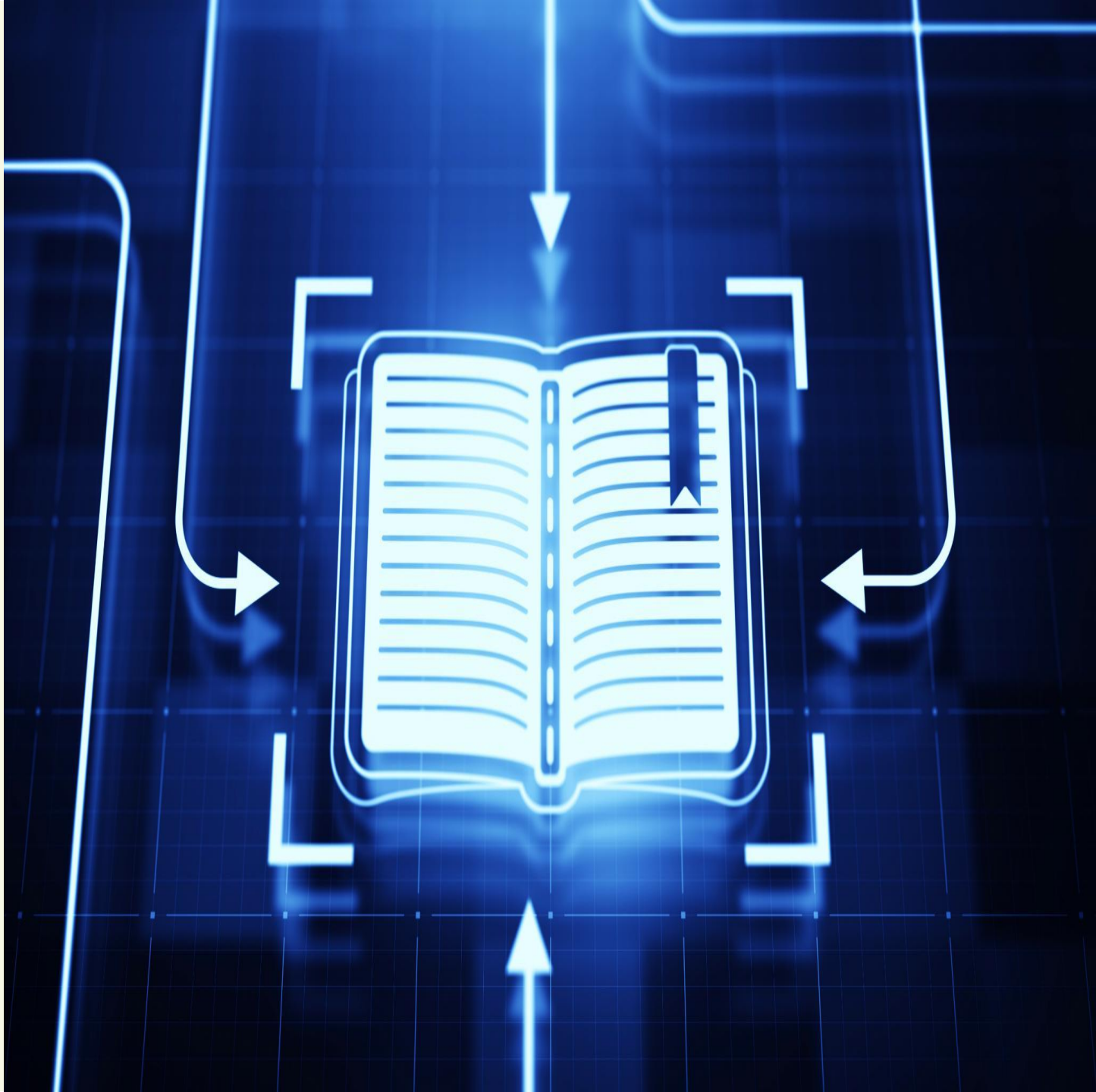
Checks availability before issuing, calculates due dates automatically, and updates loan status on return.

Validation and overdue logic

Enforces required fields and valid numeric inputs; compares dates to flag overdue items consistently.

Clean separation from UI

Implemented in Razor PageModels and repository methods, mirroring library policy while keeping behavior consistent.



Database Tables and Relationships

Database Tables & Relationships

Core structure

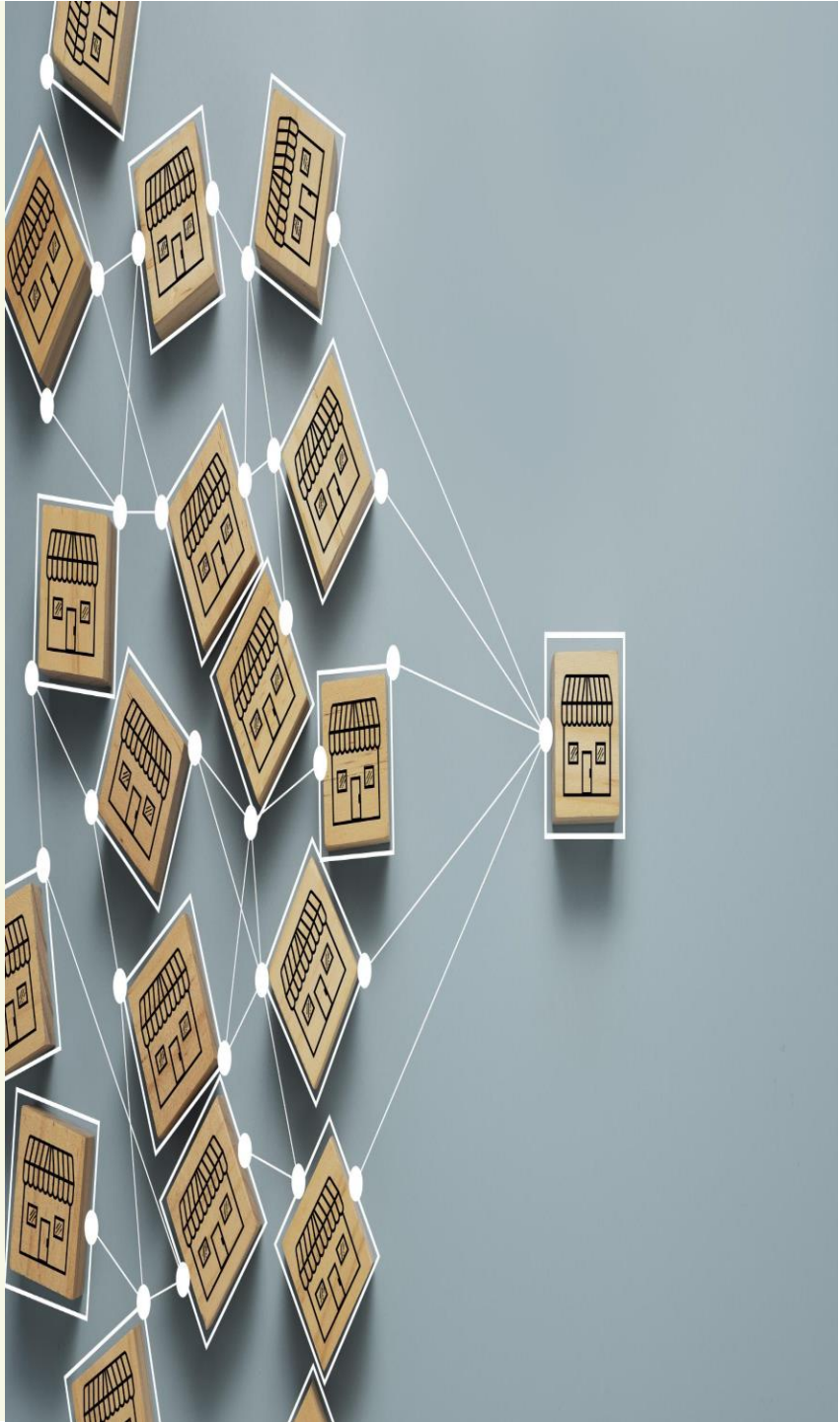
SQL Server stores application data in normalized tables: Books, Members, and Loans. Each table uses a primary key to uniquely identify records.

Relationships

Loans includes foreign keys to Books and Members, linking issued items to borrowers and enabling reliable queries for availability and loan status.

Tracking & maintainability

Timestamp fields (CreatedAt, IssuedOn, DueOn, ReturnedOn) capture key events. The design stays intentionally simple for clarity and long-term maintenance.



Key Takeaways and Conclusion

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Full CRUD Workflow

Demonstrates end-to-end CRUD using ASP.NET Core Razor Pages with SQL Server for reliable persistence.

Clean Architecture

Separates concerns across Razor views, PageModels, and a repository-based data access layer.

Real Business Rules

Implements book and member management plus loan processing with validations and clear rules.

Ready to Extend

Easy to understand and maintain, making it strong for learning, demos, and evaluation scenarios.